

The Impact of Initial Public Offerings on Rival Firms: Evidence from KOSDAQ Market

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Abstract

Our paper extends previous studies through a comprehensive examination of the impact of IPOs on the rivals. This study is guided by three primary motives: first, to investigate whether the impact of IPOs on the rivals exist, and second, to examine whether rivals' reactions in response to venture-backed IPOs are different from that of non venture-backed IPOs. Third, we also attempt to test whether the rivals react differently in response to IPOs according to the difference in specific characteristics. Results in our study show that IPOs have competitive effects on rivals. Our research also shows that venture-backed IPOs have significantly negative impacts on rivals, but rivals have insignificant reactions to non venture-backed IPOs.

Keywords : IPOs(initial public offerings), Venture Capital, Herfindahl Index

JEL Classification : G3, G14

논문투고일: 2011년 7월 31일, 수정일: 2011년 8월 10일, 게재확정일: 2011년 8월 12일

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I . Introduction

As a classic and hot issue in finance field, initial public offerings (IPOs) have attracted many researchers' attentions. In recent years, a large number of related studies have been done mainly focusing on two aspects: the underpricing of IPOs and subsequent long-run underperformance. However, relatively little evidence has been found to explain the impact of IPOs on corresponding rivals in the same industry. Thus, present research is conducted to fill the research gap.

When a company plans to go public, the news will be released to the market rapidly, and then investors may interpret valuable information revealed by an IPO as a signal of the future prospect of the firm or its corresponding industry. Therefore, it's reasonable to estimate that IPO announcements are likely to exert effects on rival firms within the same industry. However, previous studies regarding the impact of IPOs on corresponding rivals in the same industry have not achieved a consistent conclusion. Some scholars argue that a firm's decisions to go public may have competitive effects on corresponding rivals. It is because that the IPOs increase the visibility of the firm and raise funds which can be used to expand and compete more effectively, thus the newly public firms can gain more the market shares. These advantages for IPOs may exert negative effects on rivals. On the contrary, some scholars suggest that IPO announcements have contagion effects for corresponding rivals if the IPO events release favorable information about the industry prospects.

Our paper has complemented the findings of previous studies, which do not provide a clear explanation on the intra-industry valuation effect of IPO in the developing market. Considering the above-mentioned points, the key research question that motivated our work has to do with providing a new dataset from which new insight can be generated. This paper analyzes IPO firms listed on the KOSDAQ stock market, which is substantially different from the developed market. KOSDAQ is one of the fastest-growing emerging stock market and its trading volume is second only to NASDAQ market. In addition, a distinguished feature of KOSDAQ market is that the traders in the market are dominated by individual

investors rather than institutional investors. These favorable conditions provide us a solid and robust foundation for analyzing the impact of IPOs on corresponding rivals in the developing market, which has not been explored fully in the existing literature.

This study is guided by three primary motives: first, to investigate which one of the two potential externalities of IPO—competitive and contagion effect—is more powerful, and second, to examine whether rivals' reactions in response to announcements of venture-backed IPOs are different from that of non venture-backed IPOs. According to empirical evidences, venture capital firms are successful in timing the decision to take companies public (Lerner 1994) and they perform better than non venture-backed IPO firms (Brav and Gompers 1997 and Ivanov 2010). Overall, it can be concluded that venture-backed IPOs release more superior information to the market than non venture-backed IPOs, which means that, unlike non venture-backed IPOs, venture-backed IPOs should have different influences on corresponding rivals. Besides, considering the difference in the specific characteristics of IPO firms and corresponding rivals, it is necessary to test whether the rivals in the same industry react differently in response to IPO announcements. We adopt several variables, such as degree of industry concentration, book-to-market ratio, the degree of underpricing etc., to examine the direction and magnitude of rivals' reactions in response to IPO announcements.

The contributions of our research can be categorized into following dimensions: to the best of our knowledge, we are the first study to attempt to investigate whether venture-backed IPOs have different intra-industry effects on corresponding rivals compared to non venture-backed IPOs in the developing market. Second, we adopt some variables which have not been fully discussed in previous literature, such as the degree of underpricing, industry concentration and so on. Third, our research attempts to extend previous studies through a systematic and comprehensive examination of impact of IPOs on rival firms listed on KOSDAQ during the period 2006–2010. This study incorporates a comprehensive dataset of 191 IPOs firms and 3284 corresponding rivals as our research target.

The remainder of this paper is organized as follows. Section 2 reviews the literature on the intra-industry effect of IPOs on the corresponding rivals. Section 3 describes the data and methodology. Section 4 provides the results of the empirical tests, followed by the cross-sectional regression analysis in exploring possible sources of IPO intra-industry effect. The final section concludes this study.

II. Literature Review and Hypotheses Development

1. The Intra-industry Valuation Effects of IPOs

There are some studies regarding the impact of firm-level announcements on the corresponding rivals in the same industry. On the one hand, previous literature has shown that rival firms experience contagion effects if the information has industry-wide implications. Slovin, Sushka, and Bendeck (1991) find that the announcements of going-private bids generate statistically significant positive intra-industry effects. Similarly, Docking, Hirschey, and Jones (1997) detect contagion effects in response to loan loss reserve announcements. On the other hand, several scholars find that there indeed are competitive effect between firm-level announcements and corresponding rivals in the same industry. Lang and Stulz (1992) argue that bankruptcy announcements have competitive effects on rival firms. Similarly, Slovin et al. (1995) find that rivals exhibit negative reactions to the announcements of equity carve-out announcements in their sample. Other studies also verify insignificant intra-industry effects. For example, Hertzels (1991) finds there are insignificant abnormal returns for rivals in response to stock repurchases. Similarly, Howe and Shen (1998) find that dividend initiations do not contain an industry-wide implication, since the event is firm-specific with no corresponding implications for rivals. Laux, Starks and Yoon(1998) present that dividend announcements have insignificant effects on rivals. It is because that competitive effect offsets contagion effect.

Prior studies mentioned above motivate us to examine whether the IPOs have significant effects on rivals in the same industry. The rivals' reactions in response to an IPO can be either positive or negative depending on how investors make inferences about non-announcing firms' future prospects through an analysis of valuable information revealed by an IPO. It is reasonable that the rivals' reactions to IPO announcements may be the sum of these two opposing effects. In this case, net intra-industry effects of the IPO firms on corresponding rivals are decided by offsetting contagion and competitive effects.

Only a limited number of previous studies exist with regard to the impact of IPO announcements on rival firms in the same industry. In order to provide an in-depth understanding, an extended review on the impact of IPOs on rival firms is also performed. There appears to be two groups of researchers that have published empirical studies on such topic. One group of them argue that IPO announcements have competitive effects on corresponding rivals, for instance: Slovin et al. (1995) find that rivals react negatively to equity carve-out announcements, which is further interpreted as that IPOs convey unfavorable information concerning with industry prospect. Ward (1997) posits that going public, though costly, not only allows a firm to raise external capital cheaply, but also enables it to grab market share from its private competitors. Similarly, Chod and Lyandres (2011) presents that IPO announcements contribute competitive threats for their rivals. Their researches also provide several testable empirical predictions. First, going public is expected to increase an IPO firm's market share. Second, going public is expected to adversely affect the values of an IPO firm's product market rivals and cause stronger competitive effect. Maksimovic and Pichler (2001) shed light on the impact of IPO announcements on the rivals' valuation in the same industry. They explore a setting in which the going public decision conveys valuable information to competitors in the product market. By raising capital to expand or develop a new technology, IPO firms may gain strategic comparative advantages within industry, and therefore, competitors face the possibility of being displaced by a more technologically advanced rival. Stoughton et al. (2001) make a conclusion that IPO announcements have

competitive effects on corresponding rivals from the view of product quality. In addition, Akhigbe, Johnston and Madura(2006) find that the corresponding industry rivals portfolios experience unfavorable price performance on average over the 36-month period following an IPO. The dispersion in long-term industry effects following the IPOs can be partially explained by competitive effects and the timing of the IPOs.

Another group of previous studies provide different evidence regarding the impact of IPO announcements on intra-industry rival's valuation effect. Benveniste, Busaba, and Wilhelm(2002); Lowry and Schwert(2002) find an IPO, which is viewed as favorable signal for the industry as a whole, can have significant positive contagion effects on corresponding rivals in the same industry. Cotei & Farhat(2010) demonstrate that going public firms generate positive externalities creating a spillover effect for other firms to go public.

Certainly, the rivals' reactions to IPO announcements may be the sum of these two opposing effects. In this case, net intra-industry effect of the IPO firms on corresponding rivals is decided by offsetting information and competitive effects. Akhigbe, Borde and White(2003) examine the impact of initial public offerings (IPO) on rival firms and find that the valuation effects are insignificant. These insignificant reactions can be explained by offsetting information and competitive effects.

Reviewing previous studies regarding the impact of IPO announcements on rival's valuation in the same industry, we can infer that IPO announcements are likely to affect stock prices of rival firms. Thus we hypothesize:

H1 : IPOs have significant valuation effects on rival firms.

2. The Intra-industry Valuation Effects of Venture-backed IPOs and Non Venture-backed IPOs

The role of venture capitals as financial intermediaries is well documented and has been the subject of considerable research effort. After reviewing relevant literature, it is our belief that the role of venture capital in initial public offering

can be categorized into the following aspects: on the one hand, venture capital firms specialize in collecting and evaluating information of start-up and growth companies, which are more likely to be prone to informational asymmetries and capital constraints (see Megginson and Weiss, 1991; Gompers 1996). On the other hand, venture capitalists are successful in timing the decision to take the companies public (Lerner 1994). Furthermore, venture-backed IPOs benefit from the channel of financing and valuable industry expertise. Ivanov and Xie(2010) argue that venture capitalists add value to start-up firms by providing a variety of services and support that suit the specific needs of start-up operating in different industries . In the study of the Korean stock market, Yoo(2003); Lee, Kim and Ryu (2008) confirm that venture capitalists play the certification role in the process of initial public offering by analyzing the relationship between venture capitalists and IPO underpricing.

In view of the certification role of venture capital and its superior ability to collect information in Korean stock market, we can infer that venture-backed IPOs should have different impacts on rivals' valuation compared to non venture-backed IPOs. On the one hand, venture-backed IPOs are likely to offer favorable information regarding the future opportunity of the industry as a whole, which is also beneficial to the intra-industry rivals. Cotei & Farhat(2010) argue that a venture-backed IPO convey positive information about industry and this information is transferred to rival firms. Alternatively, due to the information asymmetry and the facility of financing, venture-backed IPOs can gain competitive advantages, which in turn may affect rivals adversely and cause the decrease of intra-industry rivals' valuation. Therefore, we hypothesize:

H2 : Venture-backed IPOs have different effects on rival firms within the same industry compared to non venture-backed IPOs.

3. Specific Characteristics that could influence the intra–industry valuation effects of IPOs on rivals

Considering the difference in the specific characteristics of IPO firms and

corresponding rivals, rivals may make different reactions in response to IPO announcements. Thus we adopt several variables, such as size of IPO and degree of underpricing for IPO etc., to examine the direction and magnitude of rivals' reactions in response to IPO announcements. The rationale of using these characteristics in explaining rivals' valuation effects is interpreted in detail as follows.

Market-to-book ratio: Following Akhigbe, Borde and Whyte(2003), the extent of growth opportunities within the industry, which is represented by market-to-book ratio, could affect the ability of a rival firm to resist the competitive threat from a newly public firm. Firms with relatively high market-to-book ratios are likely to have more investment opportunities than those with relatively low market-to-book ratios (Lang, Stulz, and Walkling, 1991), which means rivals with high market-to-book ratios have the ability to respond to competitive threat from IPO firms. Therefore, based on the previous studies, we hypothesis :

H3 : The intra-industry valuation effects of IPOs on rivals are negatively related to the market-to-book ratio of rival firms.

Degree of industry concentration: Compared to highly-concentrated industry, the entrance of a newly public firm in less-concentrated industry will easily bring more competitive threat for intra-industry rivals, since information conveyed by IPO announcements might signal higher product quality to consumers or investors and, reversely, non-announcing firms in the same industry are prone to lose their competitive advantage and will make negative reactions in response to IPO announcements. Thus, we argue that IPOs in less-concentrated industries are likely to reveal unfavorable information for their competitors. To test whether the degree of industry concentration for an IPO has impact on rivals' valuation, we classify rival firms into two groups: Highly-concentrated industry group and less-concentrated industry group. Highly-concentrated industry group contains the rival portfolios if the Herfindahl Index for the industry is above the median of the Herfindahl Index for all industries, vice versa. The Herfindahl Index (HI) is computed as the sum of squared market share of each firm relative to all other

firms within the four-digit SIC code. Market share is defined as the firm's annual sales at the end of fiscal year prior to the IPO announcements divided by industry sales.

H4 : The intra-industry valuation effects of IPOs on rivals are negatively related to the degree of industry concentration.

The degree of underpricing: Ward (1997) finds that when a firm announces an IPO, the stock price reactions of competitor firms are strongly negatively correlated with the IPO firm's eventual underpricing, indicating that the larger the underpricing of IPO, the more competitive threat to corresponding rivals. Thus, to test whether the degree of underpricing for IPO has different impacts on rivals, we hypothesize:

H5 : The intra-industry valuation effects of IPOs on rivals are negatively related to the degree of underpricing of IPO.

Relative sale of an IPO firm: Ward (1997) posits that an IPO not only allows a firm to raise external capital cheaply, but also enables it to grab market share from its private competitors. Therefore, it is believed that relative sale of IPO firms is an important explanatory for rivals' valuation effects. We hypothesize:

H6 : The intra-industry valuation effects of IPOs on rivals are negatively related to the relative sale of an IPO.

Relative size of rival firms: Atiase (1985) argues that firm size is significantly relative to rivals' valuation effects in response to IPO announcements. Small firms have limited ability to challenge their competitors or respond appropriately to the competitive threat. Therefore, We hypothesize:

H7 : The intra-industry valuation effects of IPOs on rivals are positively related to the relative size of rival firms.

First IPO in the industry: Akhigbe, Madura, and Whyte(2003) argue that the first intra-industry event play a vital role in rival valuation effects. Thus, we

deem that the first IPO in an industry following a period of dormancy should have significant effects on rival valuation effects. Because the advent of a newly public firm will more easily attract investor's attention to corresponding industry and future prospect. We hypothesize:

H8 : The first IPO should result in more pronounced valuation effects for rivals relative to subsequent IPO firms.

III. Data and methodology

1. Data and Sample Selection

In this study, our samples consist of 303 IPO firms listed on KOSDAQ market over the period from 2006 to 2010. Data on the listing dates, offering dates, offering prices, closing prices on the first day of trading and gross proceeds are collected directly from the official website of KRX*. Information about the proceeding of IPO and holding ratio of venture capital are collected from the system of DART**. Other data, such as market index return as well as financial statement variables of IPO firms and matching rivals in the same industry, are collected from KISVALUE database.

We exclude the following IPO firms: unit IPO, foreign IPO, and IPO not identifiable in the KISVALUE database. The final sample for industrial firms is constructed in a two-stage process described as below. In the first stage, it is required that each IPO firm has available financial information in the KISVALUE database in the first year of listing. In the second stage, the IPO sample is further reduced when we construct the sample for rival firms. In so doing, we follow three steps. First, we assign each firm for which daily stock returns are available on the KISVALUE files to a four-digit SIC code (see Lang and Stulz 1992). We exclude firms that are in their first year of listing (IPO firms). Second,

*<http://www.krx.co.kr>

**[http:// dart.fss.or.kr](http://dart.fss.or.kr).

we retain rival firms that do not have major public announcements such as mergers, seasoned equity offerings, stock splits, dividend and repurchases within a 30-day period centered at the IPO filing date. Finally, to ensure that each IPO firm is matched with a representative portfolio of rivals, we require that each IPO has at least 1 rival (same 4-digit SIC code, same year) with required financial data (total assets, sales, total liabilities) available in the KISVALUE database. The final sample is comprised of 191 IPO firms listed on the KOSDAQ and 3284 rivals in the corresponding industry, covering 65.4% of all IPO firms during the sample period. The distribution of our sample in this research is shown as table 1. As we can see from it, no considerable fluctuation in the number of IPO firms can be found during the period 2006–2010, except for a sharp decrease in the number of IPO firms in 2008. In addition, the number of venture-backed IPO firms nearly account for 65% of all the IPO firms.

〈Table 1〉 Distribution of Initial Public Offerings Listed on KOSDAQ Market During the Period 2006–2010

	2006	2007	2008	2009	2010	Total
Number of IPO firms	56	67	38	55	76	303
Number of IPO firms in Our Sample	35	47	28	44	37	191
Number of Venture-Backed IPO firms in Our Sample	21	31	21	26	24	123
Number of Non Venture-Backed IPO firms in Our Sample	14	16	7	18	13	68

2. Methodology

(1) Assessing Rival Abnormal Returns

We obtain abnormal returns (ARs) for rival firms using the standard event study methodology. The ARs are assessed on the initial trading date. Since the initial trading date is the first day on which public information becomes available about the new competition, rivals are likely to react on the initial trading date.

Following Lang and Stulz³⁾(1992)., we form an equally-weighted portfolio of publicly traded rival firms that share the same four-digit Standard Industrial Classification (SIC) code as the IPO firm on the initial trading date (day 0). We use the returns of each rival over the period (−220 to −20) relative to the IPO initial trading date to estimate the market model parameters. Cumulative abnormal returns (CARs) over three different time intervals are reported: day (0), days (0,+1), and days (0,+3). We then compute the equally-weighted average of the CARs of the rival portfolios. The abnormal return is calculated according to the risk-adjusted model using the standard one factor capital asset pricing model (CAPM) as given below:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + \varepsilon_{i,t} \quad (1)$$

$R_{i,t}$: The daily return for stock i on day t

$R_{m,t}$: The daily return for market portfolio on day t

$R_{f,t}$: The daily return for risk-free security on day t

α_i, β_i : Intercept and slope for stock i

$\varepsilon_{i,t}$: A random error term

Using the estimated parameters of α_i and β_i from equations (1), we then compute expected returns ($E(R_{i,t})$) of stock i, which, in turn, are used to measure AR and cumulative abnormal returns (CARs) for each portfolio over the post-portfolio formation period, T, as follows:

$$AR_{i,t} = R_{i,t} - E(R_{i,t}) \quad (2)$$

$AR_{i,t}$: The abnormal return for stock i on day t

3) Because of computational constraints, Lang and Stulz form industry portfolio by randomly selecting 50 firms in COMPUSTAT with the same primary four-digit SIC code as the bankrupt firm. Then they estimate abnormal returns for competitors using equally-weighted portfolio

$$AAR_t = \frac{1}{N} \sum_{i=1}^N AR_{i,t} \quad AAR_t: \text{The}$$

average abnormal return for rival portfolio on day t

$$CAR(T) = \sum_{i=1}^T AAR_t \quad (3)$$

$CAR(T)$: The cumulated abnormal return for rival portfolio within T interval (t=0,1,2,3)

(2) Cross-sectional Model

In order to examine the intra-industry rivals' different reactions in response to IPO announcements regarding IPO and rivals characteristics, we use the following model: The dependent variable is the CAR (0, 1) for each rival portfolio and the independent variables are factors that may explain the variation in rivals' valuation effects in response to IPO announcements. To explore potential sources, and trace the influence, of firm-specific characteristics on the intra-industry information effect in response to IPO announcement, we examine a series of proxies for the testable signaling hypotheses.

$$RCAR_i = \alpha_0 + \alpha_1 RELSIZE_i + \alpha_2 RELMB_i + \alpha_3 VBIPO_i + \alpha_4 DEUNDERPRING_i + \alpha_5 HI_i + \alpha_6 RELSAL_i + \alpha_7 FIRSTIPO_i + \alpha_8 TIME_i + \varepsilon_i$$

$RCAR_i$: the CAR (0, +1) of the rival portfolio

$RELSIZE_i$: the relative size of rival firms measured as the market value of the IPO firm relative to the median market value of the rival portfolio.

$VBIPO_i$: 1 if the IPO is backed by venture capital and zero otherwise.

$RELMB_i$: 1 if the market-to-book ratio of IPO exceeds industry median and zero otherwise

$DEUNDERPRICING_i$: the extent of the underpricing, P_1 is the closing price on the first trading day, and P_0 is the offer price. We estimate the underpricing as follows: $UP_i = (P_1 - P_0)/P_0$

HI_i : a dummy variable that takes on a value of one if IPO in the highly concentrated industry and zero otherwise.

$RELSALE_i$: a dummy variable that takes on a value of one if sales amount of IPO exceeds industry median and zero otherwise

$FIRSTIPO_i$: 1 if the IPO is the first in the industry within a two-year period of time prior to the current IPO; zero otherwise

$TIME_i$: 1 if the IPO occurs in 2008; 0 otherwise

ε_i : a random error term

IV. Empirical Analysis

1. The Intra-industry Valuation Effects of IPOs on Rival Firms

In Table 2, panel A shows CARs for full rival portfolios over three different time intervals CAR (0), CAR (0, 1), CAR(0,3) respectively. As seen as in panel A, the values of CAR(0), CAR(0,1) are significantly negative, it indicate that rival firms make significant negative reactions in response to IPO announcements. It can be interpreted as that the IPOs can gain competitive advantages in the Korean stock market, alternatively, competitive force of rivals in the same industry has been heavily reduced. This finding generally supports the common belief and our hypothesis regarding the impact of IPO announcements on rival

firms in the same industry. This result is also consistent with the finding of Slovin, Sushka, and Ferraro(1995).

〈Table 2〉 Cumulative Abnormal Return for Rival Portfolios

Panel A : CARs for full rival portfolios on the initial trading date of the IPO		
Event period	Number of rival portfolios	CAR(%)
(0)	191	-0.0018 ^{**}
(0,1)	191	-0.0025 [*]
(0,3)	191	-0.0033
Panel B : CARs for rival portfolios of venture-backed IPO		
Event period	Number of rival portfolios	CAR(%)
(0)	123	-0.0029 ^{***}
(0,1)	123	-0.0033 ^{**}
(0,3)	123	-0.0034
Panel C : CARs for rival portfolios of non venture-backed IPO		
Event period	Number of rival portfolios	CAR(%)
(0)	68	0.0002
(0,1)	68	-0.0011
(0,3)	68	-0.0033

***, **, and * denote significance at the 1%, 5% and 10% level, respectively.

The key research issue in our study is to examine whether venture-backed IPOs have different impacts on corresponding rivals compared to non venture-backed IPOs. We first sort all IPO firms into two groups: venture-backed IPOs and non venture-backed IPOs, base on the criteria that whether IPO firms are supported by venture capitalists or not. Panel B and panel C report CARs for rival portfolios of venture-backed IPOs and non venture-backed IPOs respectively. Our results show that venture-backed IPOs have significantly negative impacts on corresponding rivals in the same industry, but non venture-backed IPOs have insignificant effects on corresponding rivals' reactions. These results differ from those of Cotei and Farhat(2010), who find rivals make significant positive reactions in response to IPO announcements, furthermore, venture-backed IPOs also have significant positive effects on intra-industry rivals. The disparity may be

related to differences in sample size, sample composition, and time period studied.

Due to rivals' different reactions in response to IPO announcements, we divide our samples into various subgroups classified based on specific characteristics related to both IPO and rivals such as industry concentration etc. Table 3 presents CARs for rival portfolios corresponding to all selected IPOs classified based on specific characteristics of both IPO firms and rivals. In general, the results for t-statistic test are not significantly different across subgroups for most of the factors considered. However, some of the z-statistic tests are significant. As we can see from panel A, the relatively large IPOs have more pronounced effects on the corresponding rivals' valuation relative to small IPOs. In addition, the first IPOs generate significantly negative effects (-0.63%) on rivals' valuation. Rivals with relative low market-to-book ratio are more vulnerable to the competitive threats originated from IPOs announcements. We also find that rival experience significant loss when IPOs are overpriced, indicating that investors in rival firms interpret the information at the offering day as bad news for industry when IPOs are overpriced.

〈Table 3〉 CARs for Rival Portfolios Corresponding to All IPOs Classified Based on Various Characteristics

Variables	Group	CAR(0,1)	t-statistic	Number of Rival Portfolios
RELMB	1	-0.0030*	0.784	169
	0	0.0007		22
RELSIZE	1	-0.0036*	0.8840	121
	0	-0.0008		70
RELSALE	1	-0.0052	1.2510	65
	0	-0.0012		126
HI	1	-0.0017	0.5340	93
	0	-0.0033		98
FIRSTIPO	1	-0.0063*	1.0360	28
	0	-0.0019		163
DEUNDERPRICING	1	-0.0017	0.9120	139
	0	-0.0048*		52

This table shows CARs for subgroups classified based on various characteristics of the IPO, IPO firm, rivals, and the industry as a whole. The z-statistic tests the significance of the CARs and the

t-statistic tests whether the CARs are significantly different from each other. RELMB is equal to 1 if the market-to-book ratio of the IPO firm is above the median value of market-to-book value of corresponding rival portfolio, 0 otherwise; RELSIZE is equal to 1 if the market value of the IPO firm is above the median value of corresponding rival portfolio size, 0 otherwise. RELSALE is equal to 1 if the sale of the IPO firm is above the median value of sale of corresponding rival portfolio, 0 otherwise; HI is equal to 1 if the Herfindahl index for the industry is above the median of the Herfindahl Index for all industries, 0 otherwise; FIRSTIPO is equal to 1 if the IPO is the first in the industry within a two-year period of time prior to the current IPO; 0 otherwise; DEUNDERPRICING is equal to 1 if the degree of underpricing for IPO is above 0, 0 otherwise. ***, **, and * denote significance at the 1%, 5% and 10% level, respectively.

Table 4 presents CARs for rival portfolios corresponding to venture-backed IPOs and non venture-backed IPOs classified based on specific characteristics of both IPO firms and rivals. Panel A shows the cumulative abnormal returns for rival portfolios corresponding to venture-backed IPOs classified based on specific characteristics related to both IPO firms and rivals. With respect to HI, CARs are significantly different across subgroups: The IPOs with relative low Herfindahl Index have significantly negative effects on rivals' valuation, which implies that rivals in less concentrated industries are more vulnerable to the competitive threats originated from IPOs, contrast to rivals in highly concentrated industries. Along with HI, The first IPO supported by venture capitalists in an industry generates significantly negative CARs (-1%) relative to subsequent IPOs (-0.19%) and the difference is significant at the 0.05 level. These results differ from those of Akhigbe, Borde and Whyte (2003), who find rivals make significant positive reactions in response to first IPOs announcements. This disparity in results can be attributed to the different samples applied. We can infer that rivals' negative reactions in response to first IPOs mainly depend on the competitive effects of IPOs announcements rather than information effects in the Korean stock market. Table 4 also shows that rivals experience negative wealth effects in response to the announcement of venture-backed IPOs when IPOs are overpriced. As we can see from Panel C, most of cumulative abnormal returns for non venture-backed IPOs are insignificant with the exception of relative sale. This result implies that a new entrant does not represent competitive threats for corresponding rivals in the absence of venture capitalists at the time of IPOs. Overall, the results in

Table 4 are partially consistent with our hypothesis that variation in rivals' reactions can be explained by a number of different factors.

〈Table 4〉 CARs for Rival Portfolios for Venture-Backed IPOs and Non Venture-Backed IPOs Classified Based on Various Characteristics

Panel A: Venture-backed IPOs				
Variables	Group	CAR(0,1)	T-statistic	Number of Rival Portfolios
RELMB	1	-0.0026	1.577	112
	0	-0.0109**		11
RELSIZE	1	-0.0035*	0.1330	78
	0	-0.0031		45
RELSALE	1	-0.0024	0.3780	33
	0	-0.0037**		90
HI	1	-0.0005	1.8450*	61
	0	-0.0061**		62
FIRSTIPO	1	-0.0100***	2.1560**	21
	0	-0.0019		102
DEUNDERPRICING	1	-0.0028	0.544	85
	0	-0.0046**		38
Panel B: Non Venture-backed IPOs				
Variables	Group	CAR(0,1)	T-statistic	Number of Rival Portfolios
RELMB	1	-0.0038	1.336	57
	0	0.0125**		11
RELSIZE	1	-0.0037	1.0340	43
	0	0.0036		25
RELSALE	1	-0.0081	2.0340*	32
	0	0.0051		36
HI	1	0.0014	0.7980	36
	0	-0.0039		32
FIRSTIPO	1	0.0058	0.7090	7
	0	-0.0019		61
DEUNDERPRICING	1	0.0001	0.674	54
	0	-0.0055		14

This table shows CARs for subgroups classified based on various characteristics of the IPO, IPO firm, rivals, and the industry as a whole. RELMB is equal to 1 if the market-to-book ratio of the IPO firm is above the median value of market-to-book value of corresponding rival portfolio, 0 otherwise; RELSIZE is equal to 1 if the market value of the IPO firm is above the median value of corresponding rival portfolio size, 0 otherwise RELSALE is equal to 1 if the sale of the IPO firm is

above the median value of sale of corresponding rival portfolio, 0 otherwise; HI is equal to 1 if the Herfindahl index for the industry is above the median of the Herfindahl Index for all industries, 0 otherwise; FIRSTIPO is equal to 1 if the IPO is the first in the industry within a two-year period of time prior to the current IPO; 0 otherwise; DEUNDERPRICING is equal to 1 if the degree of underpricing for IPO is above zero, 0 otherwise. ***, **, and * denote significance at the 1%, 5% and 10% level, respectively.

To further explore the impact of IPO announcements on rival portfolios, CARs for rival portfolios are measured based on various cross-classifications: industry concentration and relative size, industry concentration and market-to-book ratio. As shown in table 5, panel A presents CARs for all rival portfolios. In general, the results are insignificant across different subitems in panel A, which are not consistent with our prior hypothesis that IPO announcements in less concentrated industries are likely to have significant impacts on rival portfolios compared to IPOs in highly concentrated industries. We conjecture that this disparity in results is due to integration of rivals' different reactions in response to venture-backed and non venture-backed IPOs. Aims at conducting an in-depth examination of the impact of IPO announcements on intra-industry rivals, we sort our samples into two subgroups: venture-backed IPOs and non venture-backed IPOs. Panel B shows that rivals in less concentrated industries generate significantly negative cumulative abnormal return, which means that intra-industry rivals experience wealth loss due to venture-backed IPOs announcement, regardless of the level of IPO firms and rival' size. However, the magnitude of their reactions is different. Relatively large IPOs have more pronounced influence on the corresponding rivals if they operate in less concentrated industry, compared to the relatively small IPOs in less concentrated in industry. This finding provides evidences to support our hypothesis that the entrant of the larger IPO firm is more likely to be a threat to rivals in industry with a low degree of concentration. We also find that most of the CARs related to the market-to-book ratio are insignificant with the exception of rivals with relative market-to-book ratios in highly concentrated industry, which imply that market-to-book ratio is not a determinant for rivals' valuation effect. As we can see from Panel C, cumulative abnormal returns for non venture-backed IPO are insignificant regardless of different characteristics

considered. This result implies that a new entrant does not represent a competitive threat for corresponding rivals in the absence of venture capitalists at the time of IPO.

2. Cross-sectional Results

To explore potential sources and trace the influence of IPOs on intra-industry rivals, we examine a series of proxies for the testable signaling hypotheses. Cumulative abnormal return for equally-weighted rival

〈Table 5〉 CARs for Two Cross-Classifications of Rival Portfolio

CAR(0,1)	Highly concentrated industry		Less concentrated industry	
	Mean	Numbers of rival portfolio	Mean	Numbers of rival portfolio
Panel A: All samples				
Low RELMB	-0.0027	18	-0.0002	25
High RELMB	-0.0035	80	-0.0022	68
Low RELSIZE	0.0023	32	-0.0034	38
High RELSIZE	-0.0038	61	-0.0033	60
Panel B: Venture-backed IPO				
Low RELMB	-0.0133**	10	-0.005	16
High RELMB	-0.0047	52	-0.006	45
Low RELSIZE	0.0027	21	-0.0081**	24
High RELSIZE	-0.0022	40	-0.0048*	38
Panel C: Non venture-backed IPO				
Low RELMB	0.175	8	0.0001	9
High RELMB	-0.0013	28	-0.0055	23
Low RELSIZE	0.0017	11	0.0046	14
High RELSIZE	-0.0069	21	-0.0007	22

This table shows the results for rival portfolios based on various cross-classifications. We split our samples into various subgroups based on the IPO and rivals' specific characteristics. Low RELMB group contains the rivals portfolios if the median value of market-to-book value of corresponding rival portfolio is below the market-to-book ratio of the IPO firm, vice versa; Low RELSIZE group contains the rivals portfolios if the median value of the market value of corresponding rival portfolio is below the market value of the IPO firm, vice versa. Highly concentrated industry group contains the rivals portfolios if the Herfindahl index for the industry is above the median of the Herfindahl Index for all industries, vice versa. ***, **, and * denote significance at the 1%, 5% and 10% level, respectively.

portfolios is used as a measure of information transferred from IPO firms to rivals. Model 1 shows the regression result regarding the factors related to IPO's specific characteristics. Consistent with findings in the above analysis, venture-backed IPO announcements have significantly negative impact (-0.36%) on the corresponding rivals' valuation effects. This result implies that, in terms of the certification role of venture capitalists and superior ability to collecting information, venture-backed IPOs have more pronounced competitive threats to corresponding rivals contrast to non venture-backed IPOs. In addition, the coefficient for the degree of IPO underpricing is significantly negative (-0.39%), suggesting that the stock price reactions of competitor firms are strongly negatively correlated with the IPO firm's eventual underpricing. In other word, the higher the degree of IPO underpricing is, the higher the magnitude of the rivals' negative valuation effects exists.

〈Table 6〉 Regression Results for Various Characteristics of the IPO Firms, Rival Portfolio and Industry as a Whole

	Model 1	Model 2	Model 3
INTERCEPT	0.0020	0.0024	0.0057
VBIPO	-0.0036*		-0.0045**
FIRSTIPO	-0.0017		-0.0013
DEUNDERPRICING	-0.0039*		-0.0047**
RELSIZE		0.0013*	0.0015**
RELMB		-0.0038	-0.0021
RELSALE		-0.0032	-0.0051**
HI		-0.0045**	-0.0041*
TIME		0.0006	0.0005
	R ² =0.032 Adjusted R ² =0.017 F-value=2.091* N=191	R ² =0.044 Adjusted R ² =0.018 F-value=1.732* N=191	R ² =0.084 Adjusted R ² =0.044 F-value=2.093** N=191

This table shows Regression results for various characteristics of the IPO firms, rival portfolios and industry as a whole. VBIPO is equal to 1 if the IPOs are backed by venture capital, 0 otherwise ;

FIRSTIPO is equal to 1 if the IPO is the first in the industry within a two-year period of time prior to the current IPO; 0 otherwise; DEUNDERPRICING is the degree of underpricing for IPO; RELSIZE is equal to the relative value of the market value of IPO to the median value of market value of corresponding rival portfolios RELMB is equal to 1 if the market-to-book ratio of the IPO firm is above the median value of market-to-book value of corresponding rival portfolio, 0 otherwise; RELSALE is equal to 1 if the sale of the IPO firm is above the median value of sale of corresponding rival portfolio, 0 otherwise; HI is equal to 1 if the Herfindahl Index for the industry is above the median of the Herfindahl Index for all industries, 0 otherwise.. TIME is equal to 1 if the IPO occurs in 2008; 0 otherwise***, **, and * denote significance at the 1%, 5% and 10% level, respectively.

The regression result for factors related to industry and rivals' specific characteristics is presented in Model 2. As we can see from Model 2, the degree of industry concentration is negatively related to rivals' valuation effects. This finding provides evidence to support our hypothesis that the entrant of the IPO firm is more likely to be a threat to rivals in industry with a low degree of concentration, compared to rivals in the highly concentrated industries. Consistent with our prediction that larger IPOs have more pronounced competitive effects on the corresponding rivals, the positive coefficient estimate (0.13%) of relative size is significantly at 0.1 level, indicating that the larger the relative size of IPO firms to rival portfolios, the higher the magnitude of rivals' reactions. However, the coefficient estimates for relative market-to-book ratios are statistically insignificant. One possible explanation for these results is that relative market-to-book ratios are not important determinants in interpreting the impact of IPO announcements on rivals' valuation effects.

Model 3 presents the regression result regarding specific characteristics related to both IPO firms and rivals. In general, the results are qualitatively similar to those reported when the models are estimated separately, with the exception of relative sale. Compared to model 2, the coefficient estimate for relative sale in model 3 is significantly negative (-0.51%), suggesting rivals with low sale are more vulnerable to threat from a newly public firm.

V. Conclusion

As mentioned above, the first motive of our study is to investigate whether the intra-industry valuation effects of IPOs on the corresponding rivals exist. Results in our study show that rivals have significantly negative reactions in response to IPO announcements. This finding implies that IPO firms contribute the competitive threat for rivals in the same industry, since the IPO firms not only raise funds that the firms can use to expand and compete more effectively, but also the IPO firm obtain higher market shares from its private competitors.

The second research topic in our study is to examine whether venture-backed IPOs have different impacts on corresponding rival when compared to non venture-backed IPOs. Previous empirical findings suggest that venture-backed IPO firms signal superior information to the market when compared to non venture-backed IPO firms, which means that venture-backed IPOs should have different influences on corresponding rivals when compared to non venture-backed IPOs. Our empirical findings generally support the hypothesis. By partitioning the sample into venture-backed IPOs and non venture-backed IPOs, our research shows that a venture-backed IPO have significantly negative impact on corresponding rivals in the same industry, but corresponding rivals have insignificant reaction to a non venture-backed IPO. This finding suggests that the presence of venture capitalists have constituted competitive threats for corresponding rivals in the same industry.

In addition, considering the difference in the specific characteristics of IPO firms and corresponding rivals, it is necessary to test whether the rivals in the same industry react differently in response to IPO announcements. We adopt several variables to examine the direction and magnitude of rivals' reactions in response to IPO announcements. Our results indicate that negative industry effects are more pronounced when the IPO is backed by venture capital. We also find evidence that rivals in the less concentrated industries are more sensitive to the competitive threats originated from IPO announcements. The size of IPO proceeds plays an important role in the magnitude of industry rivals' valuation effects in

response to venture-backed IPO. Rivals experience larger wealth loss in response to a larger IPO announcement in the same industry. In particular, the higher the degree of underpricing is, the higher the magnitude of the rivals' reactions exists. In conclusion, an IPO has competitive effects for corresponding rivals in the same industry.

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IPO이 경쟁기업에 미치는 영향에 관한 연구- 코스닥 시장을 중심으로

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요 약

본 연구는 IPO이 동종 산업 내에서 경쟁기업에 대한 효과에 대한 기존 선행연구들을 광범위하게 확장하려고 시도하였다. 본 연구의 목적은 다음과 같다. 첫째, IPO 이후에 경쟁기업간에 교차하여 나타날 수 있는 영향에 대해 분석하였다. 둘째, 벤처캐피탈에 투자된 IPO가 벤처캐피탈에 투자되지 않는 IPO보다 산업내 경쟁기업에 더 많은 영향을 주는지에 대해 분석하였다. 본 연구의 주요 연구결과는 다음과 같다. 첫째, IPO는 경쟁기업의 가치에 경쟁적인 영향을 줄 수 있는 것으로 나타났다. 벤처캐피탈에 투자된 IPO기업이 경쟁기업에 유의적인 부(-)의 영향을 미칠 수 있다는 결과가 나타난 것에 비해, 벤처캐피탈에 투자되지 않는 IPO기업의 경우는 유의하지 않은 결과가 나타났다.

핵심주제어 : 기업공개, 벤처캐피탈, 허핀달지수

JEL 분류기준 : G3, G14

투고일: 2011년 7월 30일, 수정일: 2011년 8월 15일, 게재확정일: 2011년 8월 21일

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